Programming Assignment #2
Campus Shuttle Flow Analyzer

Background Concepts for Network/Traffic Flow

Network analysis plays an important role in many industries such as engineering, information theory, and the study of transportation systems. The following analysis of traffic flow through a road network illustrates how systems of linear equations with many solutions can arise in practice.

Consider the typical road network of Figure 1. It represents an area of a downtown city. The streets are all one-way, with arrows indicating the direction of traffic flow. The flow of traffic in and out of the network is measured in terms of vehicles per hour (vph). We can construct a mathematical model that can be used to analyze this network.

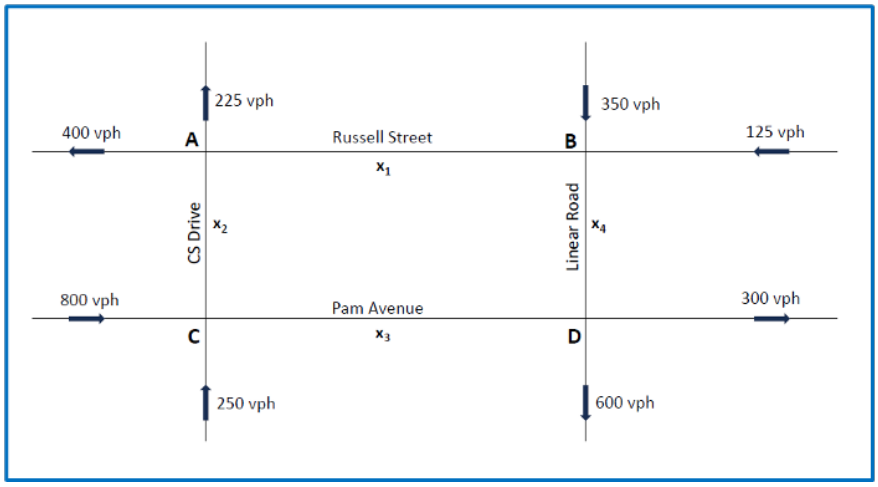


Figure 1: 4-Intersection Traffic Network

Assume the following traffic law applies:

All traffic entering an intersection must leave that intersection.

This conservation of flow constraint (similar to Kirchhoff's Current Law) leads to a system of linear equations.

Example: Building a system of equations

Intersection A: Traffic in $= x_1 + x_2$, Traffic out $= 400 + 225$, $x_1 + x_2 = 625$.

Intersection B: Traffic in $= 350 + 125$, Traffic out $= x_1 + x_4$, $x_1 + x_4 = 475$.

Intersection C: Traffic in $= 800 + 250$, Traffic out $= x_2 + x_3$, $x_2 + x_3 = 1050$.

Intersection D: Traffic in $= x_3 + x_4$, Traffic out $= 600 + 300$, $x_3 + x_4 = 900$.

These constraints on the traffic are described by the following system of linear equations:

$$\begin{aligned} x_1 + x_2 &= 625 \\ x_1 + x_4 &= 475 \\ x_2 + x_3 &= 1050 \\ x_3 + x_4 &= 900 \end{aligned}$$

The method of Gauss–Jordan elimination can be used to solve this system of equations. The augmented matrix and reduced row-echelon form of the above system are as follows.

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 625 \\ 1 & 0 & 0 & 1 & 475 \\ 0 & 1 & 1 & 0 & 1050 \\ 0 & 0 & 1 & 1 & 900 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 475 \\ 0 & 1 & 0 & -1 & 150 \\ 0 & 0 & 1 & 1 & 900 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The system of equations that corresponds to this reduced row-echelon form is

$$\begin{aligned} x_1 + x_4 &= 475 \\ x_2 - x_4 &= 150 \\ x_3 + x_4 &= 900 \end{aligned}$$

Expressing each leading variable in terms of the remaining variable, we get

$$x_1 = -x_4 + 475, \quad x_2 = x_4 + 150, \quad x_3 = -x_4 + 900.$$

As perhaps might be expected, the system of equations has many solutions (x_4 is our free variable) and therefore, many traffic flows are possible. A driver does have a certain amount of choice at intersections.

Assignment — Campus Shuttle Flow Analysis

Overview

Campus shuttle systems are often modeled using flow networks. At each shuttle stop, shuttles may enter or leave the system, and shuttles continue along assigned routes. To keep the system balanced, the number of shuttles entering each stop must equal the number of shuttles leaving that stop.

In this assignment, you will model a simplified campus shuttle route using a system of linear equations. You will construct the system, form the augmented matrix, use row reduction to find the general solution, and write a program that analyzes possible shuttle flow patterns.

Starter Code and GitHub Repository

Starter code is provided for Python, Java, and C#. Choose **one** language and complete the assignment in that language.

Use the CS2300 programming-assignment repository that you created for PA0. Add the PA2 starter files to the `ProgrammingAssignments/PA2` folder of that repository, complete your work there, commit your completed PA2, and push the submitted commit to GitHub.

The starter code contains the basic program structure, user interface, and the row-reduction routine. You are responsible for completing all portions of the network code marked `TODO`. The README file for your selected language contains the commands needed to run the program.

Important: The provided row-reduction routine must be used. Do not replace the provided routine with NumPy, SymPy, MATLAB, Mathematica, or another library/system that performs the row reduction or solves the system for you.

Campus Shuttle Network

The shuttle route contains five stops connected by six shuttle routes:

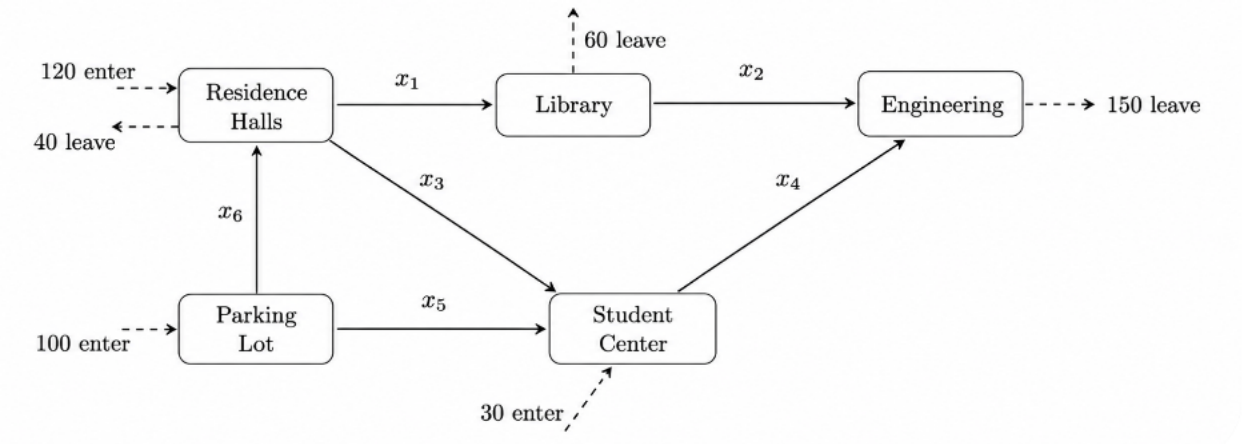


Figure 2: Campus Shuttle Network

The variables $x_1, x_2, \ldots, x_6$ represent the number of shuttles per hour traveling between campus stops.

Traffic Flow Rule

At each stop, assume the following conservation rule applies:

All shuttle flow entering a stop must leave that stop.

Assignment Tasks

You will be provided with a partially completed implementation of the **Campus Shuttle Flow Analyzer**. The provided code includes the basic program structure, user interface, row-reduction routine, and output formatting routines.

Your task is to complete the missing functionality and implement the linear algebra operations required to analyze the campus shuttle network.

Part 1: Develop the Linear System

Complete Parts 1–4 of the PA2 Mathematical Verification Form by hand before beginning the programming portion of PA2. The purpose of this work is to develop and verify the mathematical model independently before implementing it in code. Do not use your program, the provided row-reduction routine, or another computational tool to generate the equations, augmented matrix, row reduction, RREF, or general solution.

Using the campus shuttle network diagram, determine the system of linear equations that models shuttle flow throughout the campus.

Your work should clearly identify:

1. The flow entering each stop.
2. The flow leaving each stop.
3. The conservation equation for each stop.
4. The complete system of linear equations.

Part 2: Construct the Augmented Matrix

After completing the corresponding mathematical verification by hand, complete the provided code so that it constructs the augmented matrix corresponding to the shuttle flow system.

Your program should display the resulting augmented matrix in a readable format.

Part 3: Compute the General Solution

After completing the row reduction and general solution by hand on the Mathematical Verification Form, complete the provided code so that it computes the general solution of the shuttle flow system.

Your implementation should identify:

1. Pivot variables.
2. Free variables.
3. The general solution expressed in terms of the free variable(s).

Use the row-reduction routine provided in the starter code. You may not use a linear algebra library or external solver to perform the row reduction or solve the system for you.

Do not hard-code (type in) the RREF matrix, pivot/free-variable identification, solution values, or feasibility results. Obtain the RREF by calling the provided row-reduction routine on your augmented matrix; your program must compute everything else from that RREF and the user-selected free-variable value(s). The campus network itself is fixed for PA2, so the conservation equations and augmented matrix for the assigned network are constructed in the appropriate methods as part of the assignment.

Part 4: Analyze Shuttle Flow Scenarios

Complete the provided code so that it allows a user to specify a value for each free variable and generate a corresponding shuttle flow configuration.

For each scenario, your program should:

1. Compute the resulting flow values.
2. Display the flow along each shuttle route.
3. Determine whether the solution is feasible.

A shuttle flow configuration is considered feasible if all route flows are nonnegative.

Part 5: Program Output

When executed, your completed program should:

1. Display the shuttle network information.
2. Display the system of linear equations.
3. Display the augmented matrix.
4. Display the general solution.
5. Allow user-selected free variable values.
6. Display the resulting shuttle flows.
7. Indicate whether the solution is feasible.

Mathematical Verification

The **PA2 Mathematical Verification Form** is a required part of this assignment. Its purpose is to demonstrate that you understand and can independently perform the mathematics implemented by your program.

Complete Parts 1–4 of the verification form by hand before beginning the programming portion of PA2. Do not use your program, the provided row-reduction routine, or another computational tool to generate the equations, augmented matrix, row reduction, RREF, or general

solution.

After completing the program, finish the remaining portions of the verification form by using your hand-derived general solution to select values for the free variable(s), calculate the resulting shuttle flows, determine feasibility, and compare your hand calculations with the results produced by your program.

Typed or neatly handwritten mathematical work is acceptable. **The Mathematical Verification Form must include the mathematical work that supports every answer you provide.** Answers without supporting calculations, row operations, substitutions, or other required work will not receive full credit. Your work must be complete and readable enough for the grader to follow how you obtained each result.

Show how the conservation equations are formed; show the row operations and intermediate matrices used to obtain RREF; show how the general solution follows from the RREF; show the substitutions and calculations used to obtain each selected flow value; and support your feasibility conclusion using your calculated flows and the nonnegative-flow requirement.

Your completed verification form must be saved as:

`PA2_LastName_FirstName_Verification.pdf`

The completed verification PDF must be placed in your `PA2` folder, committed, and pushed to GitHub with your completed program. The commit hash submitted in Canvas must identify a commit that contains **both** your completed program and your completed Mathematical Verification PDF.

Programming and Testing Requirements

Complete PA2 using **one** of the provided languages: Python, Java, or C#.

- Use the provided row-reduction code. **Do not modify the provided row-reduction file.**
- Do not use NumPy, SymPy, MATLAB, Mathematica, or another library/system to perform the row reduction or solve the system for you.
- Do not hard-code (type in) the RREF matrix—obtain it by calling the provided routine. Likewise, do not hard-code pivot or free-variable classifications, flow values, feasibility results, or assumptions about which columns will be pivot columns or how many free variables a valid test matrix will contain.
- Your implementation must correctly perform the required computations for valid matrices/-configurations consistent with this assignment.
- You are responsible for devising and running your own tests before submission. A separate student test file is not provided.
- Code that you add or substantially modify must be appropriately documented. Include comments where needed to explain the mathematical or programming purpose of non-obvious logic. Comments are not required on every line.

Your program may be evaluated using instructor test matrices/configurations in addition to the assigned campus-network example. Instructor test matrices will use the same six route variables $x_1, \ldots, x_6$ (seven columns, including the constant column), may have a different number of rows, and will describe consistent systems. For these tests, the grader will replace the matrix returned by the augmented-matrix method; the rest of your program must work without modification.

GitHub Repository Requirements

Place your PA2 materials in the `ProgrammingAssignments/PA2` folder of the CS2300 programming repository you created in PA0. Leave the provided template structure intact and complete only the language you selected.

Before submitting in Canvas:

1. Complete and test your program.
2. Complete the Mathematical Verification Form and save it using the required filename.
3. Place the completed verification PDF in your `PA2` folder.
4. Commit all required PA2 materials.
5. Push the commit to GitHub.
6. Open the repository in a browser and verify that the commit contains your completed source code and completed verification PDF.
7. Copy the GitHub repository URL and the **full commit hash** for the exact commit you are submitting.
8. Download and electronically complete the provided `PA2_GitHub_Submission_Template.pdf`. Save it using the required Canvas submission filename.

The submitted commit identifies the version of PA2 that will be graded. Changes pushed after that commit are not part of the submitted assignment unless a new Canvas submission is made before the deadline.

Canvas Submission

Canvas is used to identify the exact GitHub version you are submitting.

Download the provided `PA2_GitHub_Submission_Template.pdf` from the PA2 assignment in Canvas. Complete the fillable PDF electronically. Do not create your own submission form.

The template requires:

- Student name
- Programming language: Python, Java, or C#
- GitHub repository URL
- Full submitted commit hash

The GitHub Submission Form must be typed in its entirety. Handwritten forms will not be accepted. Type or paste all information directly into the provided fillable fields. In particular, copy and paste the GitHub repository URL and full commit hash so they can be used to locate and verify the exact version of your work.

After completing the form, save the completed PDF using the exact filename:

`PA2_LastName_FirstName_GitHubSubmission.pdf`

Upload this completed PDF as your PA2 submission in Canvas. The Mathematical Verification PDF is **not** uploaded separately to Canvas; it must be included in the submitted GitHub commit as described above.

Required GitHub Submission: The Canvas PDF must contain a valid GitHub repository URL and the full commit hash for the exact version of PA2 being submitted. The identified commit must contain both the completed program and the completed Mathematical Verification PDF. **If the repository URL or commit hash is missing, incorrect, inaccessible, or does not identify a commit containing the required PA2 materials, the PA2 grade is zero.**

Grading Rubric

Category	**Points**	**Criteria**
Mathematical Model / System Setup	15	Correctly constructs the assigned campus-network conservation equations and augmented matrix. No hard-coded mathematical results.
Solution Analysis	20	Correctly identifies pivot and free variables and determines solution relationships from the RREF without assuming fixed pivot columns or a fixed number of free variables.
Flow Configuration & Feasibility	20	Correctly computes route-flow values from selected free-variable values and correctly determines whether a resulting configuration is feasible.
General Program Correctness & Testing	10	Program runs correctly and performs the required computations on the assigned network and valid instructor test matrices/configurations.
Mathematical Verification Form	30	Complete independent verification including conservation equations, augmented matrix, hand Gauss–Jordan row reduction (labeled row operations and intermediate matrices), RREF, pivot/free variables, general solution, hand-computed flow configuration, feasibility, and comparison with program results.
Code Quality & Documentation	5	Required starter structure is retained; code is readable; added/substantially modified code is appropriately documented; required language/folder organization is followed; prohibited solver libraries are not used.
Total	**100**	